6.3.4 Overdamped Limit of Langevin Dynamics

An interesting special case of Langevin dynamics is obtained by considering the large γ limit, often referred to as the "overdamped" limit. In cases where the friction constant is scaled by the inverse mass (so γ is replaced by γM^{-1} in (6.32)–(6.33)) such as in [233], this is instead known as the zero-mass limit. In this model the inertial dynamics is assumed to be dominated by collisional effects. Let $\boldsymbol{v} = \boldsymbol{M}^{-1}\boldsymbol{p}$ and assume that the acceleration is negligible, so that Langevin dynamics reduces to

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{v}\mathrm{d}t, \qquad 0 = -\nabla U(\boldsymbol{q})\mathrm{d}t - \gamma \boldsymbol{M}\boldsymbol{v}\mathrm{d}t + \sqrt{2\gamma \mathrm{k_B T}}\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}.$$

The second equation can be solved for $\boldsymbol{v}$ and we obtain the following SDE for $\boldsymbol{q}$:

$$\mathrm{d}\boldsymbol{q} = -\gamma^{-1}\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q})\mathrm{d}t + \sqrt{2\mathrm{k_B}\gamma^{-1}\mathrm{T}}\boldsymbol{M}^{-1/2}\mathrm{d}\mathbf{W}. \tag{6.36}$$

This is also sometimes referred to as *Brownian dynamics* or Smoluchowski dynamics. The transition from Langevin to Brownian dynamics is typically referred to as the "Kramers to Smoluchowski limit."

6.3.5 Properties of SDEs

In this section we introduce a few tools for studying the evolution of solutions for SDEs. Let X be a stochastic process satisfying the SDE (6.14), with suitable coefficients a, b independent of t, and let $\phi(\cdot)$ be any twice differentiable function. Then, taking the expectation of the Itō-Doeblin formula (6.17) we find

$$\begin{aligned}\mathbb{E}\{\mathrm{d}\phi(\mathrm{X})\} &= \mathbb{E}\{\phi'(\mathrm{X})a(\mathrm{X})\}\mathrm{d}t + \tfrac{1}{2}\mathbb{E}\{\phi''(\mathrm{X})b(\mathrm{X})^2\}\mathrm{d}t \\ &= \mathbb{E}\{(\mathcal{L}\phi)(\mathrm{X})\}\mathrm{d}t,\end{aligned} \tag{6.37}$$

where $\mathcal{L}$ is a linear operator, called the *generator* of the stochastic process, defined by

$$\mathcal{L}\phi(x) = \left(a(x)\frac{\mathrm{d}}{\mathrm{d}x} + \tfrac{1}{2}b^2(x)\frac{\mathrm{d}^2}{\mathrm{d}x^2}\right)\phi(x). \tag{6.38}$$

The generator plays a role for SDEs which is similar to that of the Lie derivative in the context of deterministic systems.